

The Spine — the chain in one pass

One continuous explanation of the aftermarket for the CFM56 (the engine that powers the 737NG and A320ceo) and FTAI Aviation's place in it, at the level of mechanism. Descriptive throughout; every figure carries its source; where two figures compete, both are given. Takes no view.

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THE SPINE

The chain in one pass

The whole chain in one reading: the engine, what it costs to keep flying, who pays, why the fleet behind the world's older narrowbodies is still flying in 2026, how that became a shop-visit plateau and a parts shortage, who sells the substitutes for new parts, and how FTAI Aviation built a module business, a capital business and a power business on top of it. Terms a reader cannot do without are defined in a box at first use; every number carries its source and date, and where two sources disagree both are given and nothing is averaged. Ten disagreements bear directly on the economics; each is presented where the chain reaches it and all are listed at the end. Each section ends with a pointer to the Part that carries the depth. Nothing is ranked or recommended.

S1. The machine: a turbofan, its modules, the life limits that force it open, and the margin that erodes

Everything in this primer follows from four physical facts about one engine. It is built in separable sections. Inside them sit parts scrapped on a clock. Its hot section loses temperature margin as it wears. And it is only as usable as its paperwork.

TURBOFAN: FAN, BYPASS AND CORE

A turbofan is a jet engine with a large fan at the front. Most of the air the fan moves passes around the hot middle, the bypass stream; a small share goes through the middle, where fuel is burned and turbines are driven. The turbines spin the fan, and the fan does most of the pushing. [Part I §1]

CFM56 (-7B AND -5B), THE 737NG AND A320CEO IT POWERS, AND THE V2500 (THE A320CEO'S ALTERNATIVE ENGINE)

Counts of CFM56s "in service" are about 23,000 (CFM website, undated), about 24,000 (Aviation Business News, Sept 2025) and more than 22,800 (Safran, end-2025); none says whether spares are included. The -7B is the Boeing 737NG's (Next Generation) only engine; the -5B powers about six in ten Airbus A320ceo (current engine option) aircraft, the rest carrying the V2500 from the IAE consortium. The last -7B was delivered by May 2019 and -5B production for airframes ended in 2022, so every CFM56 that flies tomorrow exists today. [CFM; Aviation Business News; Safran, 13 Feb 2026; GE Aerospace, 2019; ISTAT (the International Society of Transport Aircraft Trading)]

ORIGINAL EQUIPMENT MANUFACTURER (OEM), AND CFM INTERNATIONAL

The original equipment manufacturer (OEM) holds the regulator's approval of the engine's design. It is the sole source of new parts at its published catalogue price, controls the repair manual, and runs its own shops and service contracts. For the CFM56 the OEM is CFM International, the joint venture of GE Aerospace and Safran, described everywhere as 50/50: GE builds the core, Safran the fan, booster, low-pressure turbine and gearbox. The same venture makes the LEAP (CFM's new engine) for the A320neo and the 737 MAX (the re-engined 737). [CFM website; Part I §2; Part IV §25]

MODULE, AND THE CORE AS THE HOT MODULE

A module is a section of the engine built to be removed and replaced as a unit, bolted to its neighbours at flanges, with its own serial number and records. The CFM56 has three gas-path major modules, the fan and booster, the core and the low-pressure turbine (LPT), plus an accessory gearbox. The core is the hot, fast, expensive middle: the high-pressure compressor (HPC), the combustor and the high-pressure turbine (HPT). Sources count three to six modules depending on the level counted; FTAI's releases convert at three per engine ("450 modules (150 engines)", 26 Feb 2025). In FTAI's exchange business "core" also means the unit a customer hands back, called the "returned core" here. [Aircraft Commerce Issue 58; FTAI release, 26 Feb 2025; Part I §3; reconciliation R-035]

LIFE-LIMITED PART (LLP) AND THE ENGINE FLIGHT CYCLE (EFC)

An engine flight cycle (EFC) is one take-off and landing; a mature short-haul narrowbody flies roughly 1,000 to 1,500 cycles a year (Aircraft Value News). A life-limited part (LLP) is a disk, spool or shaft whose failure the engine case cannot be designed to contain. It is certified for a fixed number of cycles and must be scrapped at that limit whatever its condition: 30,000 in the fan and booster, 20,000 in the compressor and high-pressure turbine, 25,000 in the low-pressure turbine on the -7B. These are the innermost parts, so reaching a limit forces the engine open. [Aircraft Commerce Issue 34; records mini-packs; Aircraft Value News]

EXHAUST-GAS TEMPERATURE (EGT) MARGIN

Every engine has a certified exhaust-gas temperature (EGT) red line. EGT margin is the gap between that limit and the temperature the engine reaches at full take-off thrust on a hot day. As blade tips and seals wear the engine runs hotter and the margin shrinks; at zero the engine must come off, and restoring the high-pressure turbine's blades, vanes and seals recovers most of it. [Aircraft Commerce Issue 27; Part I §5]

TIME ON WING

Time on wing is the interval between an engine's installation and its removal, set by whichever clock runs out first: the life-limited part with the fewest cycles remaining or the temperature margin. A mature CFM56 has a performance restoration every 10,000 to 15,000 cycles and a full overhaul with life-limited-part replacement at 20,000 to 25,000 (Leeham, Mar 2024): roughly seven to fifteen years between restorations at 1,000 to 1,500 cycles a year. [Leeham News, 8 Mar 2024; Part I §5–§6]

WORKED EXAMPLE S.1 — THE SHORTER CLOCK GOVERNS

Inputs: a CFM56-7B whose high-pressure turbine disk has used 18,500 of its 20,000 cycles (limit from the mini-packs; cycles used assumed); 1,500 cycles a year (top of Aircraft Value News's range); temperature margin for another 5,000 cycles (assumed).

cycles remaining on the disk = $20,000 - 18,500 = 1,500$
time to the limit = $1,500 \div 1,500$ a year = about one year
margin remaining = 5,000 cycles, worth nothing to this owner

The disk sets the removal date at about a year, and 5,000 cycles of margin go unused. On a fresh disk and a worn hot section the margin governs instead, and the disk's life goes into the shop unused. Two inputs are assumed; the point is the rule. [Part I worked example 1.1]

Three consequences follow. An engine at a shop visit is really three or four assets with different remaining lives. The quickest return to service is often to replace the one module that has run out. And the usable supply is the documented supply: a module with a gap in its records cannot be installed. The fleet behind all of this only shrinks, since every engine retired, torn down or converted leaves the pool and nothing enters it. [Part I §3–§4, §9]

Depth: Part I §1–§5.

S2. What a shop visit is and costs, and why the parts bill is where the money sits

When either clock runs out, the engine comes off and goes to a shop. The bill has five lines, and the mix among them decides most of what follows.

SHOP VISIT

A shop visit is any removal of the engine for work in an engine shop, from a visit to fix one finding to a complete rebuild. The shop disassembles the engine, inspects every piece, replaces or repairs what is out of limits, rebuilds it and runs it in a test cell before release. "Teardown" is reserved here for dismantling an engine to sell its parts. [Part I §6]

MAINTENANCE, REPAIR AND OVERHAUL (MRO), AND THE INDEPENDENT MRO

MRO is the industry's generic term for maintenance, repair and overhaul work and the companies that do it. An independent MRO is an engine shop not owned by the engine's manufacturer: a stand-alone company such as StandardAero, an airline's maintenance arm, or another engine maker's shop such as MTU (a German engine maker). All work to the manufacturer's manual. [Part I §6; Part IV §25]

WORKSCOPE, PERFORMANCE RESTORATION AND OVERHAUL

The workscope specifies which modules are opened, how deep, and which parts are replaced new, repaired or swapped for used material. A performance restoration opens the core to recover temperature margin; the fan and low-pressure turbine are inspected but not fully disassembled. An overhaul disassembles every module and replaces the life-limited parts that would fall due in the next run ("zero-time": a complete new set). A hospital or quick-turn visit fixes one thing and returns the engine; a module exchange is one form of it. [Part I §6; registry C-7]

TURNAROUND TIME (TAT) AND THE INDUCTION SLOT

Turnaround time (TAT) runs from an engine's induction into a shop to its release: 90 to 120 days for a full CFM56 overhaul in 2025 against about 60 before the pandemic (trade press); "sustaining about 90 days" in GE's shops (May 2026). The induction slot is the date a shop commits to start; when shops are full the wait for one, two to six months (Bain, 2024), is additional to the turnaround, and every day away needs a spare engine or grounds an aircraft. [Aviation Business News, 2025; GE Aerospace, 27 May 2026; Safe Fly Aviation, 2026 (consultancy blog); Bain, 2024]

THE SHOP-VISIT COST BUILD

A visit's cost is labour, new OEM parts, repairs, life-limited-part replacement (new or used) and the test-cell run, plus freight, consumables and the shop's margin. Material is 60 to 70% of the total, labour 20 to 30% and repairs 10 to 20% (Air Cargo Week); a consultancy blog adds that life-limited parts are 40 to 60% of parts cost when replaced and high-pressure turbine blades and vanes 15 to 25%. [Air Cargo Week, 2025/26; Safe Fly Aviation, 2026; Part I §7]

CATALOGUE LIST PRICE AND ESCALATION

The catalogue list price is the manufacturer's published price for every new spare part, repriced at least annually; shops pass it through at list plus a handling margin, and used and alternative parts are priced as discounts to it. Escalation is the annual increase: around 10% a year on new spares in 2022 and 2023 (Aviation Week); Safran's chief executive described pricing "above inflation by 3 to 4 points"; and the published price of a full -7B life-limited-part set compounded at about 7% a year from 2008 to 2025, derived in the research. [Aviation Week, late 2023; Part I worked example 1.3; Part IV §28; reconciliation R-018]

EXHIBIT S.2 — WHAT A CFM56 SHOP VISIT COSTS: THE BINDING RANGES WITH THEIR SOURCE LEVEL

Quantity	Figure	Source and level	Note
Performance restoration	\$1.2–1.6M on one page; \$1.8–2.5M on another	Safe Fly Aviation, 2026, two pages (consultancy blog)	Part IV uses the first, Part I the second; never averaged (R-013)
Heavy visit with life-limited parts	\$2.1–2.8M; ">\$3.5M"	Safe Fly Aviation, two pages	Neither fits a full new set at current list
Full overhaul ("zero-time")	\$3.5–4.2M	Safe Fly Aviation	Full new set by the label
Full -7B life-limited-part set at list	\$1.775M (2008); \$3.4M (2018); "\$4m" (Nov 2019); \$5.7M (June 2025, scope not visible)	Aircraft Commerce Issues 58 and 120; Aircraft Value News; MyAirTrade	Part IV computes with \$4M, Part I with \$5.7M (R-016)
Cost shares	Material 60–70%, labour 20–30%, repairs 10–20%; LLPs 40–60% of parts, HPT blades and vanes 15–25%	Air Cargo Week; Safe Fly Aviation	Two secondary sources (R-014)

Source: Safe Fly Aviation, 2026 (consultancy blog); Aircraft Commerce Issues 58 and 120; Aircraft Value News, Nov 2019; MyAirTrade, June 2025; Air Cargo Week, 2025/26. Reconciliation E.2, rules 4 to 6. Accessed 2026-10-03.

Every cost figure in the exhibit is from trade press or one consultancy blog, and that blog disagrees with itself on the same scope; no manufacturer's or shop's price list was read. [reconciliation R-013] The set prices supply a check: a restoration at \$1.4M plus a full new set at \$4M is \$5.4M, above every quoted "with LLPs" range, so those totals must assume partial replacement, used parts or older prices. [Part I §7.1]

WORKED EXAMPLE S.2 — STUB LIFE, AND WHY THE LIFE-LIMITED-PART DECISION OUTWEIGHS THE WHOLE RESTORATION

Inputs: an engine arrives for a restoration at 12,000 cycles (assumed); its core parts have a 20,000-cycle limit (sourced), so 8,000 remain; the next run is planned at 12,000 cycles (assumed, inside Leeham's range). A full new set lists at \$5.7M (MyAirTrade, June 2025, scope not visible) or about \$4.0M (Aircraft Value News, Nov 2019). The restoration costs \$2.15M or \$1.4M, the midpoints of the two consultancy ranges.

keep the parts: the engine comes off again after 8,000 cycles, forced by the limit, with its restoration two-thirds used
 replace them new now: 8,000 of each part's 20,000 cycles (40%) are scrapped unused
 cost per cycle, full life used: $5,700,000 \div 20,000 = \285 (at \$4.0M: \$200)
 cost per cycle, scrapped at 12,000: $5,700,000 \div 12,000 = \475 (at \$4.0M: \$333)
 restoration alone, per cycle of the next run: $2,150,000 \div 12,000 = \179 (at \$1.4M: \$117)

The shorter remaining lives already inside the engine, the "stub life" (Aircraft Commerce), add a third to a half to the cost of every cycle if the parts are scrapped early, a penalty larger than the whole restoration on either set price. The third option, which the used-parts market supplies, is used life-limited parts with 12,000 to 15,000 cycles remaining; their prices were not retrieved. [Aircraft Commerce Issue 34; Part I worked examples 1.2 and 1.6; reconciliation R-016, R-017]

The parts bill is where the money sits for three reasons. Material is 60 to 70% of a visit and is priced off the catalogue, so escalation flows into every visit whoever performs it. Life-limited parts are 40 to 60% of parts cost when replaced and no alternative maker produces one for the CFM56, so that line is single-source. And the high-pressure turbine's blades and vanes, the next largest line, sit where most of the lost margin is recovered (S8). A shop's levers against the manufacturer are therefore all on the material line: used parts, approved alternative parts, repair instead of replacement, and exchange of a module instead of overhaul of the engine. [Part I §7; Part VI §38]

Depth: Part I §6–§7; Part IV §29.

S3. Where the money sits across the chain: a first margin map, bases stated, nothing ranked

Before the chain goes further, its links and the profit each reports give the segment sections a frame. This is a map, not an analysis.

MARGIN POOL

The margin pool is this primer's term for how the profit earned in keeping an engine type flying is distributed across the links of the chain: the manufacturer, the shops, the repairers, the alternative-parts makers, the used-material traders, the lessors and the servicers of third-party capital. No company reports a "margin pool"; the term names a way of reading the filings together. [Part I §11; Part XI §86]

FTAI'S TWO REPORTABLE SEGMENTS: AEROSPACE PRODUCTS AND AVIATION LEASING

Aerospace Products "develops and manufactures, repairs/refurbishes, and sells aircraft engines and aftermarket components primarily for the CFM56-7B, CFM56-5B, and V2500" through its Module Factory (S9); FY2025 segment revenue was \$1,936.2M. Aviation Leasing "owns and manages aircraft and engines and leases and sells them, directly and through an equity-method investment"; it held 290 assets at 31 December 2025 (S10). FTAI Power appears as a third line in the 2027 guidance; whether it is a reportable segment was not confirmed (S11). [FTAI 10-K FY2025; FTAI release, 29 Jul 2026; Part VII §45; Part VIII §58]

ADJUSTED EBITDA (CONSOLIDATED AND BY SEGMENT) AND GUIDANCE

Adjusted EBITDA is earnings before interest, taxes, depreciation and amortisation as FTAI adjusts them: items treated as non-recurring or non-cash are removed and the company's pro-rata share of Adjusted EBITDA from its equity-method investments is added back. FTAI reports it consolidated (\$597.3M in 2023, \$862.1M in 2024, \$1,190.9M in 2025) and by segment (Aerospace Products \$160.0M, \$380.6M, \$671.3M; Aviation Leasing \$500.1M and \$608.9M for 2024 and 2025), and sets guidance in the segment measure. The segments do not sum to the total (Corporate and Other about – \$89.2M in 2025). [FTAI 10-K FY2025; Part X §80, §84]

EXHIBIT S.3 — INDICATIVE MARGIN FIGURES ALONG THE CHAIN, EACH ON ITS OWN STATED BASIS

Link in the chain	Company and segment	Revenue	Margin	Basis	Source
A shop visit, by slice	Any CFM56 visit	n/a	Material 60–70%; labour 20–30%; repairs 10–20%	Share of cost, not a margin	Air Cargo Week, 2025/26
Manufacturer	GE Aerospace, Commercial Engines & Services	\$33,314M; services \$25,010M	26.6%	GE segment profit (GE's own measure)	GE 10-K FY2025
Approved alternative parts, with repair and distribution	HEICO (a maker of approved alternative parts), Flight Support Group	\$3,117.3M	24.1%	Segment operating income, derived; year to 31 Oct 2025	HEICO 10-K FY2025

Link in the chain	Company and segment	Revenue	Margin	Basis	Source
Independent engine shop	StandardAero, Engine Services	\$5,354.0M	13.2%	Segment Adjusted EBITDA, StandardAero's definition	StandardAero 10-K FY2025
Independent shop (context, non-SEC)	MTU Aero Engines, commercial maintenance	€6.0B	8.0%	Adjusted EBIT (earnings before interest and tax)	MTU, 24 Feb 2026
Teardown and used material	AerSale, Asset Management Solutions	\$211.6M	35.0%	Gross margin, derived	AerSale 10-K FY2025
Module exchange, module and engine sales	FTAI, Aerospace Products	\$1,936.2M (FY2025); \$875.0M (Q2 2026)	34.7%; 28.5%	FTAI-defined segment Adjusted EBITDA, derived; includes related-party Maintenance, Repair and Exchange (MRE) Contract revenue (S9, S10)	FTAI 10-K FY2025; Q2 2026 release
Engine and aircraft lessor	FTAI, Aviation Leasing	Lease income \$235.2M	Segment Adjusted EBITDA \$608.9M; no margin computable	Includes gains, maintenance revenue and the equity-method share	FTAI 10-K FY2025

Source: as listed in each row; compiled from Part I Exhibit 1.14 and Part XI Exhibit 11.4. Bases and fiscal years differ. Accessed 2026-10-03.

Three cautions travel with the table. The margins sit on five bases: an EBITDA margin exceeds an EBIT margin by the depreciation share of revenue, about nine points at FTAI in 2025 (\$225.8M on \$2,507.4M). A share of cost is not a margin. And a parts pass-through compresses a percentage: a shop billing \$2M of the manufacturer's parts at a small margin beside \$1M of its own labour shows a thin percentage, a parts maker a wide one, and the dollars may be similar. FTAI's segment sells a module or engine it owns, so its margin is built more like a parts business's than a shop's. [FTAI 10-K FY2025; Part X §80; Part XI §86.9]

One structural fact belongs on the map. Safran describes about 85% of the CFM56's aftermarket revenue as spare parts and time-and-materials work, against about 40% for the LEAP, and about 15% of the CFM56 fleet is under long-term flight-hour agreements against about 50% of the LEAP fleet. On this engine the manufacturer earns mainly by selling parts at list into visits it does not perform or control, which leaves room for every substitute in S8 and S9; on a contracted engine the manufacturer chooses the parts. [Safran via Aviation Week, c. 2023/24; Part IV §28]

Depth: Part I §11; Part XI §86.

S4. Who owns engines and how they pay: leases, reserves, engine values, green time

Almost every dollar an engine earns passes through a lease, and almost every dollar its owner borrows is secured on that lease. The lease is where the life an engine consumes is turned into cash, and where its value is written down part by part.

OPERATING LEASE

An operating lease rents an aircraft or engine from its owner, the lessor, to an operator, the lessee, for a fixed term and periodic rent; the lessor keeps title and the risk that the asset is worth more or less than expected when it comes back. Nearly all of FTAI's leases are operating leases. At the end of the term the lessor extends, re-leases, sells or parts out. [FTAI 10-K FY2025; Part II §12]

MAINTENANCE RESERVES

Maintenance reserves are payments the lessee makes in addition to rent, set per flight hour or cycle and collected per component (each engine's restoration and life-limited parts, the airframe's checks, the landing gear). They pre-fund the eventual shop visit; the lessor reimburses a completed event up to the balance held and keeps any unclaimed balance at lease end. FTAI books reserves it no longer has to return as "maintenance revenue" (\$218.5M in 2025). [AerCap 20-F FY2013; FTAI 10-K FY2025; Part II §13]

RETURN CONDITIONS

Return conditions are the contractual minimum state of a leased engine at redelivery: minimum hours and cycles to the next restoration, minimum cycles remaining on each life-limited part, and records in order. With the reserve regime they decide who pays for the life consumed. They may also exclude parts and repairs that are not the manufacturer's; current practice on that clause was not found. [Part II §13; Part VI §42]

LEASE RATE FACTOR (LRF)

The lease rate factor (LRF) is monthly rent divided by the asset's market value; 1.0% a month is a 12% gross annual yield before depreciation, interest and maintenance. No source states a factor for FTAI, for Willis Lease or for the CFM56; the primer's figures are derived: 0.70% a month (mixed bases), 0.87 to 1.1% (FTAI's 2025 book), about 1.0% (Willis Lease), and 0.9% assumed in one fee example. [Part II §12, §14; reconciliation R-030]

HALF-LIFE VALUE AND MAINTENANCE-ADJUSTED VALUE

Half-life is an appraisal convention, not a physical midpoint: the value of an engine with every module halfway between restorations and every life-limited part at half its life, so that no adjustment has been made for actual maintenance status (ISTAT); IBA (an appraiser) publishes CFM56 values on that basis. Maintenance-adjusted value adds, for each item, the actual remaining fraction less one half, times the cost to restore it. [ISTAT Jetrader, 2010; IBA, Sept 2025; Part I §8]

GREEN TIME

Green time is the usable life left on an engine or module before its next mandatory shop visit, set by whichever runs out first: the shortest-lived part, the temperature margin or a hard inspection. A green-time engine is one the owner will not restore, flown or leased as-is and then torn down. The pool is fed by retirements, so it shrinks exactly when a slow-retiring fleet most wants it. [Part II §16; registry C-2]

EXHIBIT S.4 — CFM56-7B VALUES AND RENTS: EACH FIGURE WITH ITS BASIS, SOURCE LEVEL AND DATE

Quantity	Figure	Basis	Source and level	Date
-7B24 and -7B27 market value	\$5.7M; \$6.4M; "levelled off after approximately 18 months of increases"	Half-life, H2 2025	IBA, Engine Values 2025B and H1 2026 (appraiser)	Sept 2025; 2026
-7B "green-time value"	\$2.8–3.4M	Labelled "(new condition)", internally inconsistent	Safe Fly Aviation (consultancy blog)	2026
Run-out -7B as feedstock	Bought \$0.8–1.2M; yields \$1.6–2.2M of used material; net \$0.4–0.8M	Exhausted engine	Safe Fly Aviation	2026
-7B spare-engine rent	~\$100,000 a month, from ~\$75,000 in 2019	Spare engine; condition not stated	IBA via AviTrader	Apr 2024

Quantity	Figure	Basis	Source and level	Date
-7B rent, alternative reading	\$42,000–48,000 a month; "about 34% above 2019"; up 15–20% "in recent months"	Beside green-time values; may be green-time	Safe Fly Aviation	2026
12-year-old 737-800 rent	~\$228,000 a month, down 11% year on year	Whole aircraft	IBA via India Seatrade News	Jul 2025

Source: IBA, Sept 2025, H1 2026 and via AviTrader, 25 Apr 2024; Safe Fly Aviation, 2026 (consultancy blog); India Seatrade News, 6 Sep 2026. Compiled from Part I Exhibit 1.11 and Part II Exhibit 2.3. Reconciliation R-028, R-029, R-031. Accessed 2026-10-03.

Two disagreements in the exhibit reach every later Part. On rent, the appraiser's \$100,000 a month and the consultancy's \$42,000 to \$48,000 agree on the rise since 2019, about a third, and disagree on the level; the lower figure sits beside green-time values and may price a part-worn engine. On value, the appraiser's half-life figures have "levelled off" while the blog's green-time value comes with rents still rising; the bases differ and so does the direction. [IBA, Apr 2024, Sept 2025, H1 2026; Safe Fly Aviation, 2026; reconciliation R-028, R-029]

Two -7B spare engines at the appraiser's rent would command \$200,000 a month, 88% of the \$228,000 a 12-year-old 737-800 earned in July 2025 (IBA); at the consultancy's rent, \$84,000 to \$96,000, 37 to 42%. The first mixes 2024 and 2025 rents and the second may price part-worn engines, so both are carried, but neither disputes the direction: the engines are separable, their value is dominated by maintenance condition, no new CFM56 is being built, and a 15-to-20-year-old airframe has few other uses. FTAI states the consequence: it "owns and leases jet aircraft which often facilitates the acquisition of engines at attractive prices". [IBA, Apr 2024 and Jul 2025; Safe Fly Aviation, 2026; FTAI Q2 2026 release; Part II worked example 2.6]

SPECIAL-PURPOSE VEHICLE (SPV), ASSET-BACKED SECURITIZATION (ABS) AND WAREHOUSE FACILITY

A special-purpose vehicle (SPV) is a bankruptcy-remote entity that owns only a defined pool of assets and the debt secured on them. An asset-backed securitization (ABS) sells leased aircraft or engines to such a vehicle, which issues notes in tranches paid in order of seniority; the sponsor usually keeps the equity and services the leases for a fee, and rating agencies size the senior notes by stressing rents, downtime, maintenance and residual values, so an appraiser's note that values have levelled off reaches the debt a pool can carry. A warehouse facility is a revolving bank loan to an SPV that funds assets until there are enough to securitize. FTAI's Strategic Capital Initiative (SCI) vehicles use all three (S10); on 22 May 2026 the first of them priced FTAI MRE 2026-1, \$612M of notes on 48 A320ceo and 737NG aircraft on lease to 23 airlines, with Series A expected to be rated Asf by Fitch and A(sf) by KBRA (a rating agency) and Series B in the BBB (triple-B) category, BBB+sf, by Fitch. [FTAI release, 22 May 2026, via Finviz and AviTrader; Part II §17]

Depth: Part II §12–§17.

S5. Why 2026 looks the way it does: the fleet that only shrinks, retirements that did not come, new engines that did not relieve, airframers that did not deliver

Nothing in S1 to S4 changes between a quiet year and a tight one; what changes is the quantities. Between 2019 and 2026 four things moved them the same way.

INSTALLED BASE, RETIREMENT RATE AND THE SHOP-VISIT FORECAST

The installed base is the population of engines fitted to aircraft in the operating fleet; the three CFM56 counts of S1 may include spare and stored engines, and a consultancy blog's about 14,200 "active" and more than 19,000 "flying" are labelled consultancy figures and never the fleet figure. The retirement rate is the share of a fleet withdrawn in a year: GE's planning assumption for 2026 moved from 3–4% of the base, to 2–3%, to 1.5–2% (GE CFO via Aviation Week, 2026); GE reported about 1.5% retired in 2025 and expected about 2% in 2026 (via Leeham, 22 Jan 2026); first-quarter 2026 retirements ran "below 1%" (GE investor-relations page). The shop-visit forecast counts expected inductions a year: 2,300 to 2,400 CFM56 visits a year in 2026 to 2028 with the apex around 2027 to 2028 and a gradual fade (GE via Aviation Week, 2026), against Safran's earlier peak of about 2,500 in 2025 to 2026 (c. 2023/24). [Part III §18, §19, §23; reconciliation R-001, R-004, R-006]

Because no new -5B or -7B is built, the base falls by exactly the number retired or converted each year, and an engine that does not retire both accumulates cycles toward the visits of S2 and withholds its parts from the used-material market: the same number sits on both sides of the aftermarket.

The fleet count is contested, and every per-engine figure in the chain divides by it: none of the three manufacturer-level figures states its basis, and the consultancy's 14,200 was read in the research both as the whole active fleet and as the -7B fleet alone. [reconciliation R-001, R-002] The retirement record is contested twice over. GE's planning number for 2026 fell three times within a year and the realised rate fell further, with the two venues differing by half a point on the 2026 expectation. At aircraft level an appraiser said fewer than 200 aircraft retired in 2024 and a data provider recorded 587, with the scope visible in neither; arithmetic on other sources gives about 410 and about 1,130 a year, and Cirium's own series ran 502, 465, 333 and 325 for 2019 to 2022. [GE CFO via Aviation Week, 2026; Leeham, 22 Jan 2026; IBA via AviTrader, 24 Sep 2024; Cirium via Informa and via Aerodynamic Advisory, IATA (the airlines' trade body) conference, 2023; reconciliation R-004, R-005]

WORKED EXAMPLE S.5 — THE GAP BETWEEN THE PLANNED AND THE REALISED RETIREMENT RATE, IN ENGINES

Inputs: a base of about 24,000 (Aviation Business News, Sept 2025) or more than 22,800 (Safran, end-2025); rates at the midpoints of GE's ranges, 3.5% for "3–4%" and 1.75% for "1.5–2%" (midpoints are the primer's choice); the 2025 actual of about 1.5% and 2026 expectation of about 2% (GE via Leeham).

on 24,000: 3.5% = 840 engines a year; 1.75% = 420; 1.5% = 360; 2.0% = 480; below 1% = fewer than 240
 on 22,800: 3.5% = 798; 1.75% = 399; 1.5% = 342; 2.0% = 456
 plan less realised: $24,000 \times (0.035 - 0.0175) = 420$ a year; over 2024–26
 about 1,200 to 1,260

About 400 engines a year on any manufacturer-level base, roughly 1,200 over three years, are still flying that the manufacturer's first plan had retired: each is a future shop visit and a teardown that did not happen. The three-year figure is an upper bound. [Part III worked example 3.2]

Why did so few retire? An aircraft is retired when its next heavy event costs more than the aircraft is worth kept, and when a replacement is available. From 2023 to 2026 the second condition failed: an airline with no new aircraft pays for the visit rather than shrink, and a lessor whose lessee cannot find a replacement extends. Cirium's count that only about a dozen mid-life 737NGs and fewer than 25 A320ceos were available to rent anywhere measures how closed the gate was. [Cirium Ascend via Bloomberg; Part III §19]

THE CONTEXT SHOCK: THE GEARED TURBOFAN (GTF) POWDER-METAL ISSUE, LEAP DURABILITY AND ITS KIT, AND THE 737 MAX RATE CAP

Three failures of replacement kept the old fleet flying. Pratt & Whitney's geared turbofan (GTF) on the A320neo had turbine discs made from contaminated metal powder, disclosed by RTX (Pratt & Whitney's parent) in 2023; 835 GTF-powered A320neos were grounded or stored in late 2025 on Cirium's count against RTX's planning figure of about 350, and aircraft-on-ground (AOG) levels were down more than 20% from the 2025 peak by January 2026. CFM's LEAP came off wing early, at 2,000 to 6,000 cycles, in hot and harsh environments; CFM certified a LEAP-1B durability kit with the Federal Aviation Administration (FAA) on 18 July 2026, with production cutover expected in early 2027. And the FAA capped 737 production at 38 a month from January 2024, raising it to 42 in October 2025 and completing the review to 47 in May 2026, against a 2018 peak of 52; Airbus delivered 793 aircraft in 2025 against a target of 820 and expects A320neo rate 75 only by the end of 2027. [Cirium via The Flying Engineer, 2026; Leeham, 27 Jan 2026; Visual Approach; CFM release, 18 Jul 2026; TravelPulse, Oct 2025; CAPA (the Centre for Aviation), May 2026; Wikipedia compilation; Argus Media; Air Data News; Part III §20–§21]

The mechanism from each shock to the CFM56 is indirect and specific. A grounded A320neo needs its flying replaced by an A320ceo on the same pilot rating, which works the -5B and the V2500 harder, not the -7B. An early LEAP removal is a LEAP visit in the manufacturer's

network, but an operator whose 737 MAX fleet is less available keeps its 737NGs, which touches the -7B. A cap of 38 a month against 52 holds back on the order of 170 737NG replacements a year at a one-for-one ceiling. Each shock carries its own reversal: the groundings were easing by January 2026, the LEAP-1B kit cuts over in early 2027, and the backlogs are owed rather than cancelled, so when rates rise the deferred retirements arrive with them. [Part III §20.3, worked example 3.4]

The lessors closed the loop. With 75 to 90% of expiries extended for four to six years, the maintenance events that would have fallen at a lease transition were paid by the incumbent airline inside the extended term, and only 10 to 25 aircraft of every 100 came back to the used market as acquisition candidates or teardown feedstock. The resulting forecast is contested between the two CFM partners, a peak of about 2,500 against a plateau of 2,300 to 2,400 through 2028, and either depends on a retirement rate that moved three times in a year. [Cirium Ascend via Bloomberg; IBA, Nov 2023; Part III worked example 3.5; reconciliation R-006, R-007]

Depth: Part III §18–§23.

S6. The translation table in one page: each driver, the requirement it creates, what it does not stress

Part III turns each documented driver into a numbered row stating the requirement it creates and what it leaves untouched or pushes the other way; a driver that raises shop-visit demand but removes teardown feedstock has done two things.

EXHIBIT S.6 — THE TRANSLATION TABLE: FOURTEEN DRIVERS, THE REQUIREMENT EACH CREATES, AND WHAT EACH DOES NOT STRESS

Row	Driver and its binding figure	Requirement it creates	What it does not stress, or pushes the other way
T-1	Low CFM56 retirements: ~1.5% in 2025, ~2% expected 2026, "below 1%" in 1Q 2026, against a 3–4% plan (GE)	More engines accruing cycles: sustains 2,300–2,400 visits a year through 2028; more LLP replacements and full restorations; module demand	Starves teardown feedstock (T-9); does not change shop capacity
T-2	GTF groundings: 835 aircraft (Cirium, late 2025); easing from the 2025 peak (RTX)	A320ceo lift: ceo extensions and short leases; V2500 and -5B visits brought forward; spare and green-time modules	No -7B demand except through general scarcity; adds no used material

Row	Driver and its binding figure	Requirement it creates	What it does not stress, or pushes the other way
T-3	LEAP durability: removals at 2,000–6,000 cycles; LEAP-1B kit certified 18 Jul 2026, cutover early 2027	LEAP visits in the manufacturer's network; indirectly, slower ceo and 737NG retirements	No CFM56 visits, modules or used material; the indirect support fades after cutover
T-4	737 MAX cap: 38 → 42 → 47 a month; 260 and 440 delivered in 2024 and 2025	737NG extensions and utilisation: -7B visits, LLP replacements, spares, green-time modules	Nothing on -5B or V2500; reverses as the cap lifts toward 52
T-5	Airbus shortfall: 793 against 820 in 2025; rate 75 only by end-2027	A320ceo extensions and utilisation: -5B and V2500 visits and spares; low ceo part-out	Does not stress the -7B; reverses as the rate rises
T-6	Lessor extensions: 75–90% of expiries; ~12 737NGs and <25 A320ceos available; terms 4–6 years	Maintenance paid by the incumbent inside the term; higher mid-life lease rates; capital held in aircraft on lease	Only 10–25 of every 100 aircraft return: fewer acquisition candidates and less feedstock; rent direction contested (R-031)
T-7	OEM escalation: ~10% a year 2022–23; inflation plus 3–4 points (Safran); -7B LLP set ~7% a year 2008–25	Demand for substitutes on the material line, 60–70% of every visit: used material, approved alternative (PMA) parts, repair, module exchange	Does not change the removal count; leaves the LLP line unsubstituted, since no PMA exists for a CFM56 LLP
T-8	Capacity shortfall: slot waits 2–6 months; overhaul 90–120 days; demand to exceed capacity by >17% by decade end (Bain)	Spare engines and leases in proportion to the delay; exchanges and quick-turns in place of overhauls	Does not change removals; GE reports turnaround improving (R-019, R-020)
T-9	Used-material scarcity: supply "remains thin" (Aviation Maintenance magazine, 2025); AerSale whole-asset sales 20 engines → 13; values up as much as 50% in two years (ePlaneAI)	Higher used-part and green-time prices; more new parts bought; PMA as the other substitute; capital to acquire feedstock	A consequence of T-1, T-4, T-5 and T-6, reversing when they do

Row	Driver and its binding figure	Requirement it creates	What it does not stress, or pushes the other way
T-10	Data-center power and the turbine shortage: data-center electricity to ~945 TWh by 2030 (International Energy Agency (IEA)); GE Vernova backlog plus reservations 116 gigawatts (GW) at Q2 2026, slots sold for 2031	A new sink for CFM56 cores: 100 Mod-1 units in 2027 would consume ~100 engines a year at one core per unit	No aviation visits and no used material; ~100 engines is 21–28% of a year's retirements at 1.5–2%; engines per unit undisclosed
T-11	Freighter conversion: >260 737-800 freighters in service, >300 in the pipeline; IBA warns of "oversupply" (6 Sep 2026)	Green-time -7B modules and visits for 20-plus-year-old airframes	Removes airframes and engines from the teardown pool; contested (R-034)
T-12	Mature-narrowbody utilisation "rising significantly" (IBA, Sept 2024)	Visits and LLP replacements earlier in calendar time; faster margin erosion	No used material; ends when replacements arrive
T-13	GTF groundings easing: AOG down 25% year to date, output up 40% (RTX, Jan 2026)	Returning neos reduce ceo lift: ceo rates soften; more -5B and V2500 teardown supply later	Does not reduce visits already induced; resolution put at 2027–28
T-14	CFM56 plateau then decline: 2,300–2,400 through 2028, apex 2027–28, gradual fade (GE); ~2,000 visits by 2030 implied (Safran)	Long run: shop capacity moves to the LEAP; teardown supply rises as retirements normalise (~2.7% a year from 2028, McKinsey)	Does not stress 2026–27 demand; timing depends on a retirement rate revised three times in a year

Source: Part III Exhibit 3.14 and the sources cited there; rows T-7, T-9 and T-10 draw on Part I §4, Part V §33 and Part IX §71. Reconciliation R-001 to R-007, R-019, R-020, R-031, R-034. Accessed 2026-10-03.

Read down the right-hand column and the chain's central tension is visible before any segment is described. The drivers that raise demand for visits, modules and spares (T-1, T-4, T-5, T-6) are the drivers that cut the supply of used parts and green-time engines (T-9), because the engine that is not retired is both. T-7 alone acts on price, raising the value of every visit and the demand for every substitute for a new part. And the one driver that removes engines for a non-aviation reason (T-10) competes with the teardown trade and the exchange business for the same run-out cores. The rows are answered by the shop-visit market (S7: T-1, T-7, T-8), the substitutes (S8: T-1, T-6, T-7, T-9, T-11), the Module Factory (S9: T-1, T-2, T-4, T-7, T-8), leasing and capital (S10: T-2, T-4, T-5, T-6, T-8, T-9) and FTAI Power (S11: T-10). [Part III §24, Exhibit 3.15]

Depth: Part III §24.

S7. The shop-visit market and the OEM's pricing: demand plateau, capacity shortage, list-price escalation, the OEM's own offerings

The market the engine creates has four kinds of shop, the manufacturer's network, the large independents, the airlines' maintenance arms and, since 2022, FTAI's Module Factory, and one price-setter, the manufacturer, which earns from nearly every visit without performing it.

Demand is a plateau whose height and length are contested. Safran's earlier framing had CFM56 visits peaking at about 2,500 a year in 2025 and 2026 and then falling; GE's 2026 framing has 2,300 to 2,400 a year, "probably closer to 2,400", held through 2028 with the apex around 2027 to 2028 and a gradual fade. The later view is lower at its top and later in reaching it; the CFM56 is about a quarter of all engine shop work by count (about 2,300 of about 9,000 visits in 2025). The shape moved with the retirement assumption, so the fade's start date is the least certain number in the chain. [Safran and GE via Aviation Week; Aviation Week 2026 Fleet & MRO Forecast via ePlaneAI; reconciliation R-006, R-007]

WORKED EXAMPLE S.7 — RECONCILING 2,300 VISITS A YEAR WITH THE INSTALLED BASE

Inputs: 2,300 visits a year (GE via Aviation Week, 2026, low end); bases of about 24,000, more than 22,800, more than 19,000 "flying" and about 14,200 "active" (the last two consultancy, labelled); a mature restoration every 10,000–15,000 cycles at 1,000–1,500 cycles a year (Leeham; Aircraft Value News), roughly 6.7 to 15 years; spares at about 10% of installed engines (MTU).

on 24,000: $2,300 \div 24,000 = 9.6\%$ a year $\rightarrow$ about 10.4 years between visits
on 22,800: 10.1% $\rightarrow$ 9.9 years; on 19,000: 12.1% $\rightarrow$ 8.3 years; on 14,200:
16.2% $\rightarrow$ 6.2 years
removing 10% spares from 24,000: $2,300 \div 21,600 = 10.6\% \rightarrow$ 9.4 years

An implied interval of about ten years matches the mature interval only at the long end of its range; about six years on the consultancy base is shorter than any sourced interval. So the 24,000 base is consistent with the sourced interval only if it includes a few thousand spare, stored or green-time engines, or if long first runs are depressing the visit rate: about 70% of the fleet had seen at most one shop visit in 2024 (Safran). Which base is right decides every per-engine figure in the chain, including FTAI's share of it. [Part III worked example 3.6; reconciliation R-001, R-008]

Capacity has two readings and both are carried. Trade press and consultancies describe a full system: overhauls at 90 to 120 days, slot waits of two to six months, demand set to exceed engine-maintenance capacity by more than 17% by the end of the decade (Bain, 2024), and two-thirds of operators short of technicians (Oliver Wyman, April 2026). GE's filing and conference material read the other way: turnaround for the LEAP, CFM56 and GE90 (a widebody engine) improved by more than 10% year on year in the fourth quarter of 2025, and CFM56 turnaround

was "sustaining about 90 days" in May 2026. The consequence does not depend on which is right: a removal that returns in five to ten months ties up a spare for that long, so the shortage of slots is itself demand for spare engines and for anything that returns the aircraft to service in weeks, which is the exchange of S9. [Aviation Business News, 2025; Bain, 2024; Oliver Wyman, Apr 2026; GE 10-K FY2025; GE Aerospace, 27 May 2026; reconciliation R-019, R-020]

The manufacturer's model on this engine has four elements. Spare parts are sold to any shop at the published catalogue price, which escalated about 10% a year in 2022 and 2023 and, for -7B parts, 20 to 30% after August 2023 on one shop's account (Magnetic via Aviation Week); the 2024 to 2026 steps were not found. About 85% of CFM56 aftermarket revenue is transactional, bought visit by visit with the operator or its shop choosing the parts, against about 40% for the LEAP; the remainder sits under long-term service agreements (LTSAs), multi-year contracts at a rate per flight hour under which the manufacturer performs or pays for the visits and chooses the shop and the parts, covering about 15% of the CFM56 fleet and targeted at 60 to 70% of the LEAP fleet by 2030 (Safran via Aviation Week). GE sells the same services under TrueChoice, with Flight Hour, Overhaul, Material and Transitions lines, the last covering green-time leases, exchanges and material buy-back; GE says "GE and CFM are the largest used serviceable material provider for their products" and calls the CFM56 "an open network ... with multiple providers who compete". And it signs materials agreements with shops: FTAI's of 22 January 2026 "secures OEM replacement part supply, thrust performance upgrades and component repair", terms undisclosed. [Aviation Week, late 2023, 2024 and c. 2023/24; GE Aerospace statements, 2026; FTAI release, 22 Jan 2026; Part IV §28; reconciliation R-009, R-018]

Put the elements together and the manufacturer's capture on this engine is plain. Most visits are bought visit by visit; within each, 60 to 70% of the bill is material; most material is bought new at list. So the manufacturer receives the largest slice of nearly every CFM56 visit without performing it, and its escalation lands on that slice whoever runs the shop. The substitutes of S8 and the exchanges of S9 therefore address the 85% of the aftermarket that is transactional, and would address far less of a fleet contracted the way the LEAP is being contracted; the manufacturer is on the substitutes' side too, through TrueChoice and the January 2026 agreement. [Part IV §28–§29]

An operator whose engine is at its life limit has four options: a full visit with new life-limited parts after the slot wait and the turnaround; a module exchange (S9); a green-time engine leased to fly out the remaining months; or a sale of the engine for its parts and the purchase of another. The last three exist because the engine is modular and a secondary market exists, and low retirements shrink the supply behind all three just when the slot shortage makes the first slow and escalation makes it expensive. [Part IV §30]

Depth: Part IV §25–§30.

S8. The substitutes for new parts: used material from teardowns, approved alternative parts (PMA) and designated-representative (DER) repairs

Three things can replace a new part from the manufacturer: a used part from a dismantled engine, a new part made by someone else under a regulator's approval, and a repair that keeps the old part in service. All three are priced against the manufacturer's list, and all three depend on a supply or an approval the manufacturer does not control.

USED SERVICEABLE MATERIAL (USM)

Used serviceable material (USM) is any part that has been in service, removed, inspected and if necessary repaired to the manual, and released on an airworthiness tag with its full history; a part with a gap in its paperwork is scrap. Used material trades at 60 to 80% of a new part's price on average and above list when the manufacturer cannot deliver (Aviation Week); a specialist trader sells repaired engine material to an airline at 70 to 75% of list (Aircraft Commerce). Because the base is the catalogue, the used price rises with every revision even if the discount does not move.

[Aviation Week, undated; Aircraft Commerce Issue 120; Part V §31, §34; reconciliation R-022]

TEARDOWN, PART-OUT AND FEEDSTOCK

Part-out is the owner's decision to dismantle a retired aircraft or engine and sell its components; teardown is the disassembly. Feedstock is the trade's word for the aircraft and engines bought to be torn down; FTAI's filings extend it to the unserviceable module a customer returns in an exchange and to engine inventory held for module production. A teardown module is valued like an engine in miniature: parts at used-material discounts, plus the life-limited parts' remaining life (life fraction times list, discounted), less the repair to make it serviceable. Supply is a lagged function of retirements. [Part V §32–§33, §36; registry C-14]

A run-out CFM56 bought for \$0.8 to \$1.2M yields \$1.6 to \$2.2M of used material (Safe Fly Aviation, 2026, consultancy blog), 25 to 39% of a half-life -7B's \$5.7M to \$6.4M (IBA, Sept 2025); no owner tears down a \$6M engine for \$2M of parts, so engines with life are sold or leased whole and only run-out engines reach the teardown market. Gross retirements on the manufacturer-level bases are roughly 340 to 480 engines a year (1.5 to 2% of 22,800 to 24,000), of which an unknown share is torn down: some are sold whole, some leased for green time first, and from 2026 some earmarked as power-generation cores (S11). [Part V worked examples 5.1 and 5.4; Part II §16]

The 2023 to 2026 shortage follows in the steps the sources describe: the retirement wave did not arrive, would-be feedstock was leased as green time or overhauled, teardown companies competed for a smaller pool, and prices rose, with CFM56 values up "as much as 50%" in two

years on one trade-press account, "a lessors' market" (IBA, April 2024), "a seller's market for used parts" (Aviation Week) and AerSale's whole-asset sales down from 20 engines in 2024 to 13 in 2025 in "a hypercompetitive feedstock environment". [ePlaneAI; IBA via AviTrader, 25 Apr 2024; Aviation Week; AerSale, 5 Mar 2026; Part V §33] FTAI sits on both sides of the trade: it buys engines and whole aircraft for their material, and it sells the used modules and parts it does not build into modules through an exclusive partnership the annual report does not name; AAR (a parts distributor and maintenance company) announced on 27 March 2025 that the two had extended their exclusive Serviceable Engine Products agreement for CFM56 material through 2030, covering more than 450 engines as trade coverage carried it; the agreement's terms were not read. [FTAI 10-K FY2025; AAR via PRNewswire, 27 Mar 2025, body as carried by ePlaneAI; Part V §33.6, §35]

PARTS MANUFACTURER APPROVAL (PMA)

A Parts Manufacturer Approval (PMA) is the FAA's combined design-and-production approval that lets a company other than the type-certificate holder legally make and sell a replacement part, with no access to the manufacturer's data. Approvals have concentrated in consumables and simple parts. A cooled, coated high-pressure turbine blade is the hard case: its life depends on an alloy, a crystal structure and a coating the applicant must reproduce and prove in an engine test programme it funds itself. Chromalloy's approval of a CFM56-5B/7B blade on 30 October 2025, "the only aftermarket alternative to the OEM manufactured new part", is treated in the trade as that milestone. No PMA exists for any CFM56 life-limited part. [eCFR, 14 CFR (Code of Federal Regulations) Part 21 Subpart K; HEICO 10-K FY2025; Chromalloy release, 30 Oct 2025; Part VI §37–§38, §41; reconciliation R-024]

DESIGNATED ENGINEERING REPRESENTATIVE (DER) REPAIR

A Designated Engineering Representative (DER) is an engineer appointed by the FAA to approve engineering data on its behalf; a DER repair is a major repair done to data a DER approved rather than to the manufacturer's manual, so the part stays the manufacturer's part and what the manual would scrap is restored. The cadence decides which lever applies: at a first visit, around 20,000 cycles, airlines typically repair the turbine blades; at a second, around 30,000, they cannot be repaired again and a new part is needed, which is where an approved alternative blade competes. [FAA Order 8110.37F; In Practise, undated; Part VI §40]

EXHIBIT S.8 — THE SUBSTITUTES: WHAT EACH REPLACES, HOW IT IS PRICED, WHO SELLS IT, AND WHAT THE RESEARCH COULD AND COULD NOT SOURCE

Substitute	Replaces and pricing basis	Who sells it	Binding figures; not retrieved
Used serviceable material	New OEM parts, and LLPs with cycles remaining; 60–80% of new; 70–75% of list from a trader; LLPs at life fraction × list, discounted	Teardown traders (AerSale and others); lessors' part-outs; the manufacturer (TrueChoice); FTAI through its AAR partnership	Run-out CFM56 \$0.8–1.2M → \$1.6–2.2M of material; 2025 market \$5.87B, \$7.7B or \$8.55B by three vendors. Not retrieved: used LLP package prices; engines torn down a year
PMA parts	New OEM blades, vanes, seals, consumables; never a disk, spool or shaft; a discount to list no source discloses	HEICO (300–500 or 400–550 approvals a year); Chromalloy, with FTAI as partner	HPT blades and vanes 15–25% of parts cost; Chromalloy's CFM56 roster is three announced approvals by one reading, four parts by another. Not retrieved: the discount; the saving per visit
DER repair	A new part, by keeping the old one in service; repair price against list	Turbine-part repair houses; airline engineering; independents; the manufacturer (about 400 CFM56 repairs industrialised since early 2023)	Repairs are 10–20% of a visit. Not retrieved: a repair price for any CFM56 hot-section part
FTAI–Chromalloy joint venture	Hot-section parts in FTAI's own modules; "at cost" per investor blogs, not in filings	FTAI "develops and manufactures PMA parts through a joint venture" (10-K)	Blogs claim "up to \$2 million per shop visit"; FTAI's only quantified lever is "up to \$75,000" a visit from its Bristol accessories venture. Not retrieved: partner, ownership and accounting; the saving per module

Source: Aviation Week, undated; Aircraft Commerce Issue 120; Safe Fly Aviation, 2026 (consultancy blog); three market-research vendors as in Part V §35; HEICO 10-K FY2025; Chromalloy, 30 Oct 2025; CFM International; FTAI 10-K FY2025; Insider Monkey, Q3 2025 call; Komodo Capital and Kairos Research (investor blogs). Compiled from Parts V and VI. Reconciliation R-021 to R-027. Accessed 2026-10-03.

The alternative-parts lever touches only the on-condition parts line, since life-limited parts remain single-source and labour, test and margin are untouched, and it applies only at visits

where blades are scrapped rather than repaired, which the interview places at second and later visits, exactly the visits a fleet kept past its planned retirement undergoes. The claimed saving is contested at three source levels: investor blogs say "up to \$2 million per shop visit"; a set of 80 blades priced from an interview is \$1.6M; the consultancy's share puts blades and vanes at \$0.11 to \$0.63M a visit; and FTAI's one quantified saving is \$75,000 a visit from a different venture. Against it stand CFM's TRUEngine designation and lessors' manufacturer-only return conditions; an appraiser records that approved alternatives "continue to undermine full-life maintenance adjustments". [Komodo Capital; Kairos Research; In Practise; Safe Fly Aviation, 2026; Insider Monkey, Oct 2025; Aircraft Value News, undated; Part VI §42, §44; reconciliation R-015, R-026, R-027]

Depth: Part V §31–§36; Part VI §37–§44.

S9. FTAI's answer: the Module Factory and the Aerospace Products segment

FTAI's module business is the fourth lever of S2 organised as a production line. It rebuilds modules before any customer engine arrives, holds them in inventory, and sells them into engines whose one expired module would otherwise send the whole engine through a 90-to-120-day visit after a two-to-six-month wait.

MODULE FACTORY

The Module Factory is FTAI's name for its CFM56-7B and -5B modular repair and refurbishment programme, first described in 2022 at a 500,000-square-foot Montréal facility "with a capacity for up to 300 shop visits per year" and bought for \$170.0M on 9 September 2024. It is a place, a programme, and a process that builds and holds an inventory of serviceable modules for exchange, fed by returned cores, engines retired from FTAI's leasing fleet and the capital vehicles, and engines and aircraft bought for the purpose. [FTAI 10-K FY2025; FTAI release, 9 Sep 2024; Part VII §46]

MODULE EXCHANGE

A module exchange separates a customer's engine at the flanges, removes the module that triggered the removal, installs a serviceable module from inventory, tests the engine and returns it. The customer pays an exchange fee and gives up the unserviceable module, the returned core, which becomes FTAI's feedstock. The industry prices an exchange as the fee plus the difference in life-limited-part life between the modules swapped. Under US revenue rules the returned core is noncash consideration at fair value, so FTAI books revenue equal to the fee plus the core's value and carries the core in inventory until it is sold again. No provider's exchange fee, FTAI's included, was found in any source. [Part VII §46–§47; reconciliation cluster K-01]

MAINTENANCE, REPAIR AND EXCHANGE (MRE), AND MRE CONTRACT REVENUE

MRE is FTAI's name for its offer, a contrast with MRO: instead of restoring a customer's engine while the customer waits, FTAI swaps in a serviceable module or engine and takes the unserviceable unit as feedstock. It is sold as engine, module and parts sales on the "Aerospace products revenue" line (\$455.0M, \$1,079.8M and \$1,600.5M for 2023 to 2025), not split further in any retrieved filing. MRE Contract revenue is a second line, new in 2025, for sales to the vehicles of the 2025 Partnership that FTAI manages and part-owns: \$335.8M for 2025 and \$221.2M, \$182.8M and \$404.0M for the first quarter, second quarter and half of 2026. Two filings define it differently: the Q3 2025 10-Q describes "stand-ready obligations to provide replacement CFM56-5B and CFM56-7B engines to customers as they become unserviceable", and the Q2 2026 10-Q "the transaction price for sales of CFM56-5B, CFM56-7B and V2500 engines and related modules to the SPVs of the first SCI partnership"; one is an obligation and the other a sale, and both are carried. [FTAI 10-K FY2025; FTAI 10-Qs Q3 2025 and Q2 2026; Part VII §45, §48; reconciliation R-044]

EXHIBIT S.9 — AEROSPACE PRODUCTS BY PERIOD: REVENUE, THE RELATED-PARTY LINE, ADJUSTED EBITDA, MARGIN AND MODULES

Period	Aerospace products revenue	MRE Contract revenue	Segment revenue	Segment Adjusted EBITDA	Margin (derived)	Modules produced
FY2023	\$455.0M	\$0	\$455.0M	\$160.0M	35.2%	not retrieved
FY2024	\$1,079.8M	\$0	\$1,079.8M	\$380.6M	35.2%	Q4 ≈136 (derived)
FY2025	\$1,600.5M	\$335.8M	\$1,936.2M	\$671.3M	34.7%; Q4 stated 35%	757 (target 750); Q2 184; Q4 228
Q1 2026	\$522.6M	\$221.2M	\$743.8M	\$222.6M	29.9%	≈270 (derived)
Q2 2026	\$692.2M (derived)	\$182.8M	\$875.0M	\$249.7M	28.5%	296
1H 2026	\$1,214.8M	\$404.0M	≈\$1,618.8M	≈\$472.3M	≈29.2%	566
2026 guidance				\$1,050M (25 Feb 2026, reaffirmed 29 Jul 2026)	"can approach 40%" (call, 26 Feb 2026)	1,050 (Feb release); raised to 1,200 on the July call (call coverage)
2027 guidance				\$1,400M (29 Jul 2026)		

Source: FTAI 10-K FY2025; 10-Qs Q1 and Q2 2026; earnings releases 29 Jul 2025 to 29 Jul 2026; call coverage (Motley Fool, MarketBeat, GuruFocus, Nasdaq.com). Derived figures are arithmetic on filed numbers. Compiled from Part VII Exhibits 7.4 to 7.18. Reconciliation B-1, R-036, R-043. Accessed 2026-10-03.

Four tensions sit in that table. Segment revenue equals the products line plus the MRE line by arithmetic (\$1,936,244 thousand is exactly \$1,600,456 plus \$335,788 thousand); the explicit segment-note mapping was not retrieved. [reconciliation B-1, R-043] The 2026 module target has two figures: 1,050 was set on 25 February 2026, a 39% increase on 757, and coverage of the 30 July 2026 call reported it raised to 1,200; the July release text was not found to carry 1,200. [FTAI release, 25 Feb 2026; GuruFocus, Nasdaq.com and MarketBeat, 30 Jul 2026; reconciliation R-036] Cost of sales for 2025 has two figures in one filing: \$1,240,368 thousand in the segment expenses table and \$1,349,719 thousand on the consolidated line, which also carries leased assets sold; the segment figure gives a derived gross margin of 35.9%. [FTAI 10-K FY2025; reconciliation R-045] And the margin fell from 35% in the fourth quarter of 2025 to about 29 to 30% in the first half of 2026, when the related-party share of revenue was highest (17.3% for 2025; 29.7%, 20.9% and 25.0% for the first quarter, the second and the half); the research recorded the two side by side and did not infer a cause, since the filings disclose neither margin by line nor the elimination's composition (S10). [Part VII §48, §53]

Capacity is stated in four units at four dates for different footprints: "up to 300 shop visits per year" for Montréal (2022 language); 1,350 module overhauls and more than 500 engine tests a year for Montréal plus Miami (release, 9 Sep 2024); 1,800 modules and 600 engines after Rome Fiumicino was added (release, 26 Feb 2025); and 3,000 modules a year across all sites and partners (call coverage, July 2026). FTAI's releases convert at three modules per engine; Part I counts four major modules including the gearbox, so 1,200 modules is 400 engines on one convention and 300 on the other. [FTAI 10-K FY2025; FTAI releases, 9 Sep 2024 and 26 Feb 2025; GuruFocus and Quartr, Jul 2026; reconciliation R-035, R-037, R-041]

WORKED EXAMPLE S.9 — REVENUE AND ADJUSTED EBITDA PER MODULE: A BOUND, NOT A PRICE

Inputs: products revenue of \$1,600.5M (FY2025), \$522.6M (Q1 2026) and \$692.2M (Q2 2026, derived); segment revenue of \$1,936.2M, \$743.8M and \$875.0M; segment Adjusted EBITDA of \$671.3M, \$222.6M and \$249.7M; modules of 757, about 270 (derived) and 296; 2026 guidance of \$1,050M over targets of 1,050 and 1,200 modules.

products revenue per module: FY2025 $1,600.5 \div 757 = \$2.11\text{M}$; Q1 2026 $\$1.94\text{M}$; Q2 2026 $692.2 \div 296 = \$2.34\text{M}$
 segment revenue per module: FY2025 $\$2.56\text{M}$; Q2 2026 $875.0 \div 296 = \$2.96\text{M}$
 Adjusted EBITDA per module: FY2025 $671.3 \div 757 = \$0.89\text{M}$; Q1 2026 $\$0.82\text{M}$; Q2 2026 $\$0.84\text{M}$
 2026 guidance per target module: $1,050 \div 1,050 = \$1.00\text{M}$; $1,050 \div 1,200 = \$0.875\text{M}$
 second half of 2026: $1,050 - 566 = 484$ modules, 242 a quarter against 296 in Q2; for 1,200: 317 a quarter

For every module produced the segment booked about \$2M of product revenue and earned about \$0.8 to \$0.9M of Adjusted EBITDA; with the related-party line, revenue per module is \$2.6 to \$3.0M. Five limits make each an upper bound: the numerator includes whole engines, parts, used material and V2500 work while the denominator counts only CFM56 modules; a module produced is not a module sold; the profit on related-party sales is counted before elimination; exchange revenue includes the fair value of returned cores, which is not cash; and the core-to-fan mix is undisclosed.

[Part VII §56; reconciliation R-046]

The segment's share of its market is framed three ways on three denominators: about 5% of a \$22B addressable market as of 2024 (release, 26 Feb 2025); a goal of 25% of CFM56 shop visits, year unstated (call, July 2026); and, derived, about 17% of a 2,350-visit year at 1,200 modules and three per engine. Management's three levers for moving the margin toward 40% in 2026 are all on the material line of S2: the approved alternative blade, more used material and the CFM materials agreement; the first half ran at about 29%. [FTAI release, 26 Feb 2025; GuruFocus, Jul 2026; MarketBeat and Motley Fool, 26 Feb 2026; Part VII §49, §52, §57; reconciliation R-042]

THE MUDDY WATERS SHORT REPORT AND THE AUDIT COMMITTEE REVIEW

Muddy Waters Research, a short seller, published a report on FTAI on 15 January 2025 alleging, as third-party summaries and the later complaint describe it, that whole-engine sales were presented as module sales, that margins were inflated and that sales were pushed to intermediaries ahead of demand; the report itself was not read. FTAI said it "strongly disagrees"; on 18 January 2025 its Audit Committee "determined to begin a review, which included engaging independent advisors", and in February 2025 announced that "the allegations made against the Company are without merit". A securities class action in the Southern District of New York, No. 25-cv-00541, covers 23 July 2024 to 21 January 2025; the defendants' motion to dismiss, filed 20 November 2025, was pending at the research date. The committee engaged a new auditor, KPMG (an audit firm), replacing Ernst & Young LLP, effective 17 June 2025. [FTAI 8-Ks, Jan and Feb 2025; FTAI 10-Q Q3 2025; FTAI 10-K FY2025; Benzinga, 15 Jan 2025; law-firm case pages; Part VII §54; reconciliation R-063, R-064, R-066]

The definitional point behind the allegations is one the reader can now state. Three modules built into an engine and sold outright is an engine sale; the same three installed one at a time into three customers' engines are three exchanges. The revenue may be similar, and the filings present engines, modules and parts on one line, so a reader cannot tell how much is whole engines. That is the boundary the dispute concerns; nothing in the primer resolves it. [Part VII §47.4]

Depth: Part VII §45–§57.

S10. FTAI's capital: Aviation Leasing, the Strategic Capital vehicles, the money that flows between them, and the 2026 securitization

The other half of FTAI owns things. Aviation Leasing held 290 assets at 31 December 2025, 47 aircraft and 243 engines, with utilization of about 77% on the company's equity-weighted definition. Its 2025 lines were lease income of \$235.2M, maintenance revenue of \$218.5M and asset sales of \$106.9M against segment Adjusted EBITDA of \$608.9M, so rent cannot account for the earnings; maintenance revenue, asset-sale margins, a \$46.4M gain on sales to the first capital vehicle and the equity-method share must. [FTAI 10-K FY2025; Part VIII §58] The aircraft is bought for its two engines (S4), so the book is both rent and a feedstock pipeline for S9; in 2025 the company began selling it into vehicles funded by others while keeping three roles in them.

STRATEGIC CAPITAL INITIATIVE (SCI): THE 2025 PARTNERSHIP, THE 2026 SPV AND THE SEED ASSETS

The Strategic Capital Initiative (SCI), launched 30 December 2024, raises third-party capital into closed-end vehicles that buy on-lease mid-life 737NG and A320ceo aircraft; FTAI manages each as Servicer, co-invests as a limited partner, and supplies every engine shop visit for the fleet through exchanges with its MRE business. The 2025 Partnership closed on 27 October 2025 at a \$2.0B equity hard cap with a \$2.5B asset-level debt commitment, and by 17 August 2026 had "approximately \$6.0 billion of total capital across over 300 aircraft", fully committed and in "harvest phase". The Seed Assets are the 46 on-lease aircraft FTAI sold from its own book to start it, "at an estimated net purchase price of \$549 million". The 2026 SPV closed a \$2.0B warehouse facility with an accordion to \$3.0B on 14 August 2026; its equity and FTAI's stake were undisclosed. [FTAI releases, 30 Dec 2024, 26 Feb 2025, 27 Oct 2025, 17 Aug 2026; FTAI 10-K FY2025; Part VIII §60–§62; reconciliation R-055]

SERVICER AND SERVICING FEE

The Servicer manages a vehicle's assets for a fee: leasing and remarketing the aircraft, collecting rent and reserves, overseeing condition and selling. FTAI "as the Servicer, provides aircraft management services to the 2025 Partnership, and receives customary, market-based compensation": \$10.2M for 2025, \$6.99M for the second quarter of 2026 and \$12.85M for the first half. The fee's rate, its base and the income-statement line it sits on are not disclosed. [FTAI 10-K FY2025; FTAI 10-Q Q2 2026; Part VIII §61, §63]

EQUITY METHOD AND PROFIT ELIMINATION

Under the equity method an investor with significant influence but not control carries its stake at cost plus its share of earnings, less distributions and eliminated profit, on one line, with none of the investee's revenue, assets or debt consolidated. FTAI's stake in the 2025 Partnership is 19% in the FY2025 10-K and the 2026 10-Qs and 20% in the Q3 2025 10-Q. Profit elimination removes from current earnings the profit on sales to such an investee and recognises it later: "the profit from the MRE Contract revenue is eliminated through equity method earnings and will be recognized over time as the 2025 Partnership generates income". Eliminations were \$10.0M in the first quarter of 2026 and \$6.6M in the second, \$16.6M for the half; whether that is 19% of the profit or all of it is not stated. [FTAI 10-Qs Q3 2025, Q1 and Q2 2026; FTAI 10-K FY2025; Part VIII §63–§64; reconciliation R-047, R-048]

ASSET-LIGHT MODEL AND HARVEST PHASE

"Asset-light" is management's term for owning fewer leased assets directly and earning servicing fees, maintenance work and equity income from vehicles that own them; the July 2026 cut to Leasing guidance was explained as "reflecting a continued shift to an asset-light business model". Harvest phase is the period after a closed-end vehicle has finished buying, in which it collects rent, pays distributions and sells assets until it winds up; the 2025 Partnership's term and exit route were not located. [FTAI releases, 29 Jul 2026 and 17 Aug 2026; FTAI 10-Q Q2 2026; Part VIII §60–§61; reconciliation R-058]

EXHIBIT S.10 — THE STREAMS BETWEEN FTAI AND THE 2025 PARTNERSHIP, BY PERIOD, AND WHERE EACH LANDS

Stream	FY2025	Q1 2026	Q2 2026	1H 2026	Where it lands
Gain on aircraft sold to the Partnership	\$46.4M (the three 2025 quarters sum to \$50.1M on 45 aircraft)	n/l	n/l	n/l	A gain line outside revenue, under topic 610-20 of the US Accounting Standards Codification (ASC), described as "non-recurring"; inside Leasing Adjusted EBITDA

Stream	FY2025	Q1 2026	Q2 2026	1H 2026	Where it lands
Servicing fees	\$10.2M	\$5.9M (derived)	\$7.0M	\$12.85M	Line not confirmed
MRE Contract revenue, sales to and from the Partnership	\$335.8M	\$221.2M	\$182.8M	\$404.0M	MRE Contract revenue line, Aerospace Products (S9)
Profit elimination on those sales	n/l	\$10.0M	\$6.6M	\$16.6M	Through equity-method earnings
Carrying value; stake	\$281.7M; 19% (20% in Q3 2025)	\$279.4M; 19%	\$365.5M; 19%		Equity-method investment, balance sheet
Leasing guidance for 2026	\$575M (25 Feb 2026)		\$475M (29 Jul 2026)		Segment Adjusted EBITDA; 2027 guidance \$450M

Source: FTAI 10-K FY2025; 10-Qs Q1 2025 to Q2 2026; releases of 25 Feb 2026 and 29 Jul 2026. n/l = not located. Compiled from Part VIII Exhibits 8.6, 8.10 and 8.11. Reconciliation R-047 to R-049, R-054, R-057. Accessed 2026-10-03.

Three disagreements in the table bear directly on the economics. The stake is 19% in the later filings and 20% in the Q3 2025 10-Q; the research suggests dilution at the final close and marks that as a guess. The 2026 Leasing guidance was \$575M on 25 February 2026 and \$475M on 29 July 2026; a search reading of the July release that returned \$575M was established on follow-up as a mis-read of the February figure, so the path is \$575M cut to \$475M, with \$450M for 2027. And the profit elimination supports two readings of one sentence. If FTAI eliminates only the share of profit corresponding to its stake, the usual treatment, then \$16.6M implies about \$87M of gross profit on \$404.0M of first-half sales to the Partnership, a margin of about 22% (at 20%, \$83.0M); if it defers all of the profit, the gross profit was \$16.6M and the margin about 4%. The filing supports either, so the segment's margin on Partnership volume cannot be compared with its margin on third-party volume; the related-party pricing of aircraft, exchanges and the fee is not disclosed. [FTAI 10-Q Q2 2026; FTAI releases, 25 Feb 2026 and 29 Jul 2026; reconciliation R-047, R-048, R-049, R-057]

WORKED EXAMPLE S.10 — WHAT A \$6.0B VEHICLE PAYS ITS MANAGER: THE DISCLOSED RUN-RATE AGAINST THREE PEER SCALES

Inputs: servicing fees of \$6.988M for Q2 2026 and \$12.849M for 1H 2026 (10-Q); total capital about \$6.0B and equity \$2.0B (releases); a lease rate factor of 0.9% a month (assumed; alternatives 0.70% and 1.0%, S4); peer fee scales: Carlyle's aviation securitizations at about 2% of rents plus incentives

up to 5% of rents and 3% of sale proceeds; Willis Lease's engine trusts at 13.5% of net rents; a private-fund convention of 1.5% of committed equity (assumed, not an SCI term).

disclosed run-rate: $6.988 \times 4 = \$27.95\text{M}$ a year; $12.849 \times 2 = \$25.70\text{M}$
on total capital: $27.95 \div 6,000 = 0.47\%$; on equity: $27.95 \div 2,000 = 1.40\%$

rent at 0.9% a month on \$6.0B: $0.009 \times 12 \times 6,000 = \648M a year (at
0.70%: \$504M; at 1.0%: \$720M)

Carlyle scale: servicing 2% of rents = \$13.0M; incentive at the 5% cap =
\$32.4M; 3% on an assumed \$600M of disposals = \$18.0M

private-fund scale: $1.5\% \times 2,000 = \$30.0\text{M}$; Willis trust scale, for
contrast: $13.5\% \times 648 = \$87.5\text{M}$

The disclosed run-rate of \$26 to \$28M sits between the servicing-only securitization figure of about \$13M and a 1.5% management fee of \$30M, and well below the Willis scale. The placement rests on assumed inputs; FTAI's share of earnings is allocated "based on the contractual profit-sharing arrangement", and whether a promote exists is undisclosed. [FTAI 10-Q Q2 2026; Carlyle 10-K FY2025; Willis Lease 10-Ks FY2019 and FY2020; Part VIII worked example 8.5; reconciliation R-030]

The maintenance stream is the one the releases describe but never size. A 300-aircraft fleet carries 600 engines; on the intervals of S1 that is 40 to 100 removals a year, a few hundred million dollars of exchanges at the per-module bounds of S9. The disclosed sales to the Partnership, \$404.0M in the first half of 2026 alone, sit above every case built that way; the record offers four candidates without choosing: the line covers purchases as well as sales, aircraft may have arrived with engines already due, whole engines may have been sold to equip the fleet, and the per-module figures are bounds. [Part VIII worked example 8.6] The vehicle's borrowing, the February 2025 commitment, the \$612M securitization of May 2026 and whatever bank debt remains, never reaches FTAI's balance sheet; FTAI's own \$3.45B of long-term debt is its senior notes (S12). [Part VIII §61, §64] The WestJet transaction of 28 September 2026 shows the structure in one deal: 17 737-700s on lease went to the 2026 SPV, and 10 off-lease aircraft went to FTAI for their 20 CFM56-7B engines; terms were not disclosed. [FTAI release, 28 Sep 2026; Part VIII §67]

Depth: Part VIII §58–§67; Part X §82.

S11. FTAI Power: a second life for cores, the J&F order, the market it sells into

The newest business removes CFM56 engines from aviation for good. It sits on the same side of the ledger as retirement and teardown, and competes with both for the same run-out cores.

FTAI POWER: MOD-1, HEAT RATE, BEHIND-THE-METER BRIDGING POWER, AND J&F POWER SYSTEMS

FTAI Power, launched 30 December 2025, converts CFM56 engines into aeroderivative generator sets: the fan is removed and replaced by a small compressor, a power turbine and generator are fitted, and the fuel system is changed for gas. Mod-1 is its 25 megawatt (MW) trailer-mounted unit, described in call coverage as 35 to 40% efficient at a heat rate (fuel energy per kilowatt-hour) of about 9,000 and as giving "near-fully depreciated" engines a 10-to-20-year second life; the release-level statements are capacity for "over 100 units annually", first delivery by the fourth quarter of 2026 and 100 units in 2027. Behind-the-meter (BTM) generation is power made on the customer's side of the utility meter; bridging power is temporary on-site generation until a grid connection arrives. J&F Power Systems, the venture with Jereh Group (a Chinese maker of trailer-mounted gas-turbine packages) that packages and sells Mod-1, is "FTAI's joint venture" to FTAI and "its subsidiary" in Jereh's exchange disclosure as reported; the ownership split and consolidating party are undisclosed. On 22 July 2026 J&F received a five-year master supply agreement and an initial \$1.465 billion purchase order from "a leading international cloud service provider" (FTAI's release) or "a U.S. hyperscaler" (call summaries), for delivery in batches through November 2027; the unit count was not disclosed. [FTAI releases, 30 Dec 2025 and 22 Jul 2026; FTAI 10-Q Q2 2026; TheValueist and Aviation Week, call coverage, 26 Feb 2026; FilingReader, 22 Jul 2026; Part IX §68–§71; reconciliation R-061]

The market is defined by a shortage the sources date. The International Energy Agency expects global data-center electricity use to more than double to about 945 TWh by 2030. GE Vernova's gas-turbine backlog plus slot reservations rose from 50 GW in 2025 to 116 GW at the second quarter of 2026, with slots sold for 2031 delivery, and Siemens Energy quotes lead times of three years and more. A buyer that needs power in 2026 or 2027 cannot get a new frame turbine; that is the opening for a 25 MW unit built from a used airline engine, as for piston plants and fuel cells. The 2027 guidance puts FTAI Power's Adjusted EBITDA at \$450M in the release text; one call summary gives "\$450 million to \$750 million", and both are carried. [IEA, Apr 2025; Power Engineering, Jul 2026; Oilprice, 2026; FTAI release, 29 Jul 2026; GuruFocus, 30 Jul 2026; Part IX §71; reconciliation R-062]

WORKED EXAMPLE S.11 — THE IMPLIED PRICE OF A MOD-1, AND THE CFM56 CORES A HUNDRED OF THEM CONSUME

Inputs: order value \$1.465 billion (release, 22 Jul 2026); 2027 target 100 units (release, 25 Feb 2026); unit count in the order undisclosed, so 100, 80 and 60 assumed; 25 MW per unit; one gas generator per unit (assumed, by analogy with the LM2500 (GE's aeroderivative of the same class)); bases and rates as in S5; three modules per engine.

at 100 units: $1,465 \div 100 = \$14.65\text{M a unit} = \586 per kW ; at 80: $\$18.3\text{M}, \$733/\text{kW}$; at 60: $\$24.4\text{M}, \$977/\text{kW}$
 2027 Power Adjusted EBITDA per unit at 100 units: $450 \div 100 = \$4.5\text{M}$ (at $\$750\text{M}: \7.5M)
 cores at 100 units: 100 engines = 0.42% of 24,000; 21% (at 2%) to 29% (at 1.5%) of a year's retirements on OEM-level bases; 35% on $14,200 \times 2\% = 284$
 module-equivalents: $100 \times 3 = 300$ a year = 29% of the 1,050 target, 25% of 1,200, 40% of 2025's 757

The implied price sits below the \$2,000 to \$2,500 per kW of plant-level filings and near Wood Mackenzie's \$600 per kW turbine-only projection for end-2027, on differing scopes. Every line scales with the undisclosed unit count, and the margin line holds only if FTAI consolidates J&F, which is undisclosed. The cores line joins this business to the chain: a hundred engines a year is under half a percent of the base and a fifth to a third of a year's retirements, so the business competes with the teardown trade and the exchange business for the same run-out engines. [Part IX §74; reconciliation R-001, R-004, R-035, R-061, R-062]

Depth: Part IX §68–§74.

S12. The company, its numbers and what remains unknown

INTERNALIZATION, AND THE COMPANY IN ONE PARAGRAPH

FTAI Aviation Ltd. is a Cayman Islands company listed on Nasdaq. It began as Fortress Transportation and Infrastructure Investors LLC, listed on the New York Stock Exchange on 15 May 2015 and externally managed by a Fortress Investment Group subsidiary; it spun off its infrastructure businesses on 1 August 2022 and became a Cayman parent on 10 November 2022. Internalization is the termination of an external management agreement, the hiring of the manager's people, and a payment for the fee stream given up; FTAI's closed on 28 May 2024 for a stated \$300.0M, \$150.0M in cash paid on 17 June 2024 and 1,866,949 shares valued at \$150.0M, so expense lines before and after mid-2024 are not like for like. [FTAI 10-Q Q3 2024; FTAI 8-K, 28 May 2024; FTAI Infrastructure 10-K FY2024; Part X §75–§77]

The 2025 consolidated statement shows the chain's money in one place. Revenues were \$2,507.4M, of which Aerospace products \$1,600.5M, MRE Contract \$335.8M, lease income \$235.2M, maintenance revenue \$218.5M and asset sales \$106.9M; cost of sales was \$1,349.7M, depreciation and amortisation \$225.8M and interest expense \$247.8M; consolidated Adjusted EBITDA was \$1,190.9M. The company reported net income in 2024 and 2025 and consumed cash in operations in both years (+\$129.0M in 2023, −\$188.0M in 2024, −\$310.7M in 2025), because buying assets to sell is an operating outflow while buying assets to lease is an investing

outflow: a factory holding modules ahead of demand shows its purchases as operating cash out until they are sold. Inventory stood at \$1,364.3M at 31 March 2026, about a year of 2025's consolidated cost of sales. [FTAI 10-K FY2025; FTAI 10-Q Q1 2026; Part X §79–§81]

The balance sheet carries five series of senior unsecured notes due 2028 to 2033 at coupons of 5.50% to 7.875%, long-term debt net of \$3,451.1M and cash of \$412.2M at 31 March 2026, an undrawn revolver maturing 22 May 2027, and perpetual preferred shares (Series D at 9.50%, \$65.0M, remains; Series C's redemption was announced on 15 May 2026). Credit ratings and covenants were not retrieved; about \$3.0B of net debt is 2.6 to 2.7 times an annualised quarter of Adjusted EBITDA, an illustration only. The dividend rose at four consecutive results, from \$0.30 a quarter to \$0.35 (27 Oct 2025), \$0.40 (25 Feb 2026), \$0.45 (29 Apr 2026) and \$0.50 (29 Jul 2026); a \$500M repurchase program was announced on 15 September 2026. On 6 March 2026 Nicholas McAleese was appointed Chief Financial Officer, succeeding Eun (Angela) Nam. [FTAI 10-K FY2025; FTAI 10-Q Q1 2026; GlobeNewswire release titles; FTAI release, 6 Mar 2026; Part X §82–§83]

Guidance is given in segment Adjusted EBITDA, before Corporate and Other. For 2025 it was \$1.1 to \$1.15B on 30 December 2024, and both segments delivered above the range. For 2026 it moved from \$1.4B, to \$1.525B (\$1.0B and \$525M, 27 October 2025), to \$1.625B (\$1.05B and \$575M, 25 February 2026), and back to \$1.525B when Leasing was cut to \$475M on 29 July 2026. For 2027 it is \$2.3B: Aerospace Products \$1.4B, FTAI Power \$450M, Aviation Leasing \$450M. [FTAI releases, 30 Dec 2024, 27 Oct 2025, 25 Feb 2026, 29 Jul 2026; Part X §84]

EXHIBIT S.12 — THE BINDING NUMBERS OF THE CHAIN IN ONE TABLE, EACH AS THE RESEARCH FIXED IT

Quantity	Binding presentation	Source
CFM56 in service	~23,000 (CFM, undated); ~24,000 (Aviation Business News, Sept 2025); >22,800 (Safran, end-2025); >35,000 delivered; consultancy ~14,200 and >19,000 labelled	CFM; Aviation Business News; Safran; Safe Fly
Life limits and intervals, -7B	Fan 30,000; HPC and HPT 20,000; LPT 25,000 cycles; 1,000–1,500 cycles a year; restoration every 10,000–15,000, overhaul at 20,000–25,000	Mini-packs; Aircraft Value News; Leeham
Full -7B LLP set at list	\$1.775M (2008); \$3.4M (2018); "\$4m" (Nov 2019); \$5.7M (Jun 2025); ~7% a year compounded (derived)	Aircraft Commerce; Aircraft Value News; MyAirTrade

Quantity	Binding presentation	Source
Shop-visit cost and shares	Restoration \$1.2–1.6M and \$1.8–2.5M; heavy with LLPs \$2.1–2.8M and ">\$3.5M"; overhaul \$3.5–4.2M; material 60–70%, labour 20–30%, repairs 10–20%; LLPs 40–60% of parts, HPT blades and vanes 15–25%	Safe Fly; Air Cargo Week
Turnaround and escalation	90–120 days overhaul (trade press); ~90 sustaining (GE); slot wait 2–6 months (Bain); spares ~10% a year 2022–23 (Aviation Week)	Aviation Business News; GE; Bain; Aviation Week
-7B value and rent	Half-life \$5.7M / \$6.4M (IBA, Sept 2025); green-time \$2.8–3.4M, run-out \$0.8–1.2M (Safe Fly); rent ~\$100,000 (IBA, 2024) and \$42,000–48,000 (Safe Fly, 2026)	IBA; Safe Fly
Used-material price	60–80% of new; 70–75% of list from a trader; above list when the OEM is constrained	Aviation Week; Aircraft Commerce
Retirement rate and visits	GE planning 3–4% → 2–3% → 1.5–2%; 2025 ~1.5%; 2026 ~2%; 1Q 2026 "below 1%"; visits 2,300–2,400 in 2026–28 (GE) against ~2,500 peak 2025–26 (Safran); just above 2,000 in 2019	Aviation Week; Leeham; GE investor-relations page
OEM model	~85% of CFM56 aftermarket transactional vs ~40% LEAP; ~15% vs ~50% of fleets under long-term agreements	Safran via Aviation Week
Modules and capacity	757 (FY2025, target 750); ~270 (Q1 2026); 296 (Q2 2026); 566 (1H 2026); target 1,050 raised to 1,200 (call coverage); capacity 300 visits / 1,350 / 1,800 modules (600 engines) / 3,000 (call coverage); three modules per engine	FTAI releases and calls
Aerospace Products	Revenue \$455.0M / \$1,079.8M / \$1,936.2M (2023–25, incl. MRE \$335.8M); Adjusted EBITDA \$160.0M / \$380.6M / \$671.3M; Q2 2026 \$875.0M and \$249.7M; cost of sales FY2025 \$1,240.4M (segment) and \$1,349.7M (consolidated); MRE Contract revenue \$404.0M (1H 2026)	FTAI 10-K FY2025; 10-Qs; releases
Aviation Leasing and SCI	290 assets (47 aircraft, 243 engines), ~77% utilization; Adjusted EBITDA \$500.1M / \$608.9M; guidance \$575M → \$475M; 2025 Partnership \$2.0B equity, \$2.5B debt, ~\$6.0B total, 300+ aircraft; seed 46 for ~\$549M; stake 19% / 20%; fees \$10.2M and \$12.85M; eliminations \$16.6M; 2026 SPV \$2.0B warehouse; ABS \$612M on 48 aircraft	FTAI 10-K; 10-Qs; releases
FTAI Power	Launched 30 Dec 2025; Mod-1 25 MW; >100 units a year; first delivery Q4 2026; 100 units 2027; J&F \$1.465B (22 Jul 2026); 2027 Adjusted EBITDA \$450M (">\$450–750M" in one summary)	FTAI releases; call coverage

Quantity	Binding presentation	Source
Consolidated and corporate	2025 revenues \$2,507.4M; Adjusted EBITDA \$1,190.9M; operating cash flow −\$310.7M; inventory \$1,364.3M, long-term debt \$3,451.1M, cash \$412.2M (31 Mar 2026); guidance 2026 \$1.525B, 2027 \$2.3B; internalization 28 May 2024, \$300.0M; dividend \$0.30 → \$0.50	FTAI filings and releases

Source: reconciliation section E.2 and the orchestrator's resolutions, as presented in the sections above; each figure's underlying source is cited where it first appears in this Spine. Accessed 2026-10-03.

Read backwards from this table, the chain is one sentence long. A company that owns aircraft bought for their engines, that rebuilds the modules of an engine no longer in production into exchanges for a fleet that did not retire on schedule, that sells those exchanges in part to vehicles it manages and part-owns, and that has begun turning run-out cores into generators for a power market short of turbines, depends on three quantities: how many CFM56 engines there are, how fast they leave, and what a new part from their only manufacturer costs. Each is contested, and the ten disagreements below are where the contest sits.

Depth: Part X §75–§85; Part XII §94.

Tensions carried in the Spine

Id	The two (or more) figures	Where in the Spine
R-001 / R-002	CFM56 in service: ~23,000 (CFM); ~24,000 (Aviation Business News); >22,800 (Safran); consultancy ~14,200 and >19,000 (Safe Fly)	S1 box; S5 box and text; S7 worked example; S8; S11
R-004 / R-005	GE 3–4% → 2–3% → 1.5–2%; 2025 ~1.5%, 2026 ~2%; 1Q 2026 "below 1%"; 2024 aircraft retirements "fewer than 200" (IBA) against 587 (Cirium)	S5 box, text and worked example S.5; S8; S11
R-013	Restoration \$1.2–1.6M and \$1.8–2.5M; heavy with LLPs \$2.1–2.8M and ">\$3.5M" (one consultancy blog, two pages)	S2, Exhibit S.2, worked example S.2
R-016 / R-017	Full -7B LLP set \$5.7M (MyAirTrade, Jun 2025) or ~\$4M (Aircraft Value News, Nov 2019); \$285 or \$200 a cycle	S2, Exhibit S.2, worked example S.2
R-028 / R-029	-7B rent ~\$100,000/month (IBA) against \$42,000–48,000 (Safe Fly, may be green-time); value \$5.7M/\$6.4M "levelled off" (IBA) against green-time \$2.8–3.4M and rising rents (Safe Fly)	S4, Exhibit S.4; S8
R-035 / R-037	Four major modules (anatomy) against three per engine (FTAI releases); capacity 300 visits / 1,350 / 1,800 / 3,000 modules in different units, footprints and dates	S1 box; S9; S11

Id	The two (or more) figures	Where in the Spine
R-036	2026 module target 1,050 (release, 25 Feb 2026) against 1,200 (call coverage, 30 Jul 2026)	S9, Exhibit S.9, worked example S.9
R-043 / R-044	Segment revenue = products + MRE by arithmetic, mapping not retrieved; MRE Contract revenue as "stand-ready obligations" (Q3 2025 10-Q) against "the transaction price for sales" (Q2 2026 10-Q)	S9 box, Exhibit S.9
R-045	Cost of sales FY2025 \$1,240,368K (segment table) against \$1,349,719K (consolidated line); 35.9% derived	S9; S12
R-047 / R-048 / R-049	Eliminations of \$16.6M imply ~\$87M of gross profit if proportional to 19% or \$16.6M if fully deferred; stake 19% against 20%; Leasing guidance \$575M cut to \$475M (a \$575M July reading was a mis-read)	S10, Exhibit S.10

Unknowns carried in the Spine

The register of everything the research could not find is Part XII §94. Nineteen clusters group the entries that ask one question; the five largest are these. K-02, appraiser values for CFM56 engines (five entries): no half-life base, -5B, full-life or mid-life aircraft value was reached, so every valuation example stands on IBA's two -7B values. K-01, module-exchange pricing (three entries): no provider's exchange fee was found, so every per-module figure is a bound. K-17, segment figures by year (three entries): the 2023 Leasing lines, the Adjusted EBITDA reconciliation and net income by year. K-05, a stated PMA discount to list (three entries): neither Chromalloy's nor HEICO's, so the alternative-parts saving cannot be sized. K-18, the capital tied up in module inventory (two entries): the inventory change and the investing lines of the cash-flow statement. Also open, because S10 turns on them: the servicing fee's rate and base (K-09) and the elimination method.

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